Modelling Cross-Country Innovation Performance Using Economic Indicators and Machine Learning

I. Introduction

Background and Context

Innovation is recognised as a critical driver of economic growth which in turn, drives social change (Ding, 2022; Damanpour & Schneider, 2006). Policymakers, scholars and economists alike have researched the drivers of innovation to better understand which policies result in greater rates of national innovation (Ding, 2022; Bloom *et al.*, 2019; Mowery, 1998). Innovation can be defined as, a continuous and dynamic process where new or improved ideas, products, or services, create practical value or disrupt existing structures (Taylor, 2017).

To compare innovation across economies, the Global Innovation Index (GII) has become a widely recognised benchmark, which scores and ranks each country's innovation outputs by analysing institutions, human capital, infrastructure, business activity, and knowledge outputs (WIPO, 2024). While the GII provides a comprehensive ranking, it remains unclear which economic indicators can reliably predict innovation performance.

Research Question

To what extent can a country's innovation outcome, proxied by the Global Innovation Index (GII), be predicted from economic indicators, and which variables best explain cross-country differences in innovation outcomes?

II. Literature Review

Given the strategic importance of innovation for global progress, governments are placing it at the centre of their growth strategies (United Kingdom. Department for Business, Energy and Industrial Strategy, 2021).

Endogenous growth theory argues that economic development is principally driven by internal economic activity such as investment in human capital (Romer, 1994). Building on this, Mazzucato (2016), Mazzucato and Dibb (2019), and Deleidi *et al.* (2019) argue that strategic public investments can shape markets and can drive long-term innovation. Accordingly, indicators including gross domestic product (GDP), gross national income (GNI), and

government final consumption expenditure not only reflect economic development but also actively influence national innovation ecosystems (World Bank Group, n.d.).

Additionally, trade openness, population size, and household consumption provide insights into demand-side dynamics that may encourage private sector innovation, aligning with demand-driven theories of technical progress (Shabbir, Liyong and Iftikhar, 2021; OECD, 2011). Meanwhile, exchange rates and value-added metrics are critical indicators of a country's global competitiveness and structural economic capacity (Furman *et al.*, 2002). Stable and favourable exchange rates can encourage firms to invest in research and development (R&D), while high value-added production may foster innovation by reflecting technological intensity and skilled labour in production (Alfaro *et al.*, 2018; Hsu *et al.*, 2022; Ivanova *et al.*, 2016).

Raghupathi and Raghupathi (2017) examined how economic development indicators predicted technology sector patents in OECD countries between 2000 and 2010. Using IBM Cognos, results demonstrated that lower GDP countries had a higher share of foreign patent owners, while higher levels of education were positively associated with innovation.

Malik (2020) examined macroeconomic and institutional determinants of innovation across 15 Asian countries between 2008-2017, identifying institutional quality, education, and trade activity as key drivers using patent applications per capita and the Generalised Method of Moments technique method.

While past studies offer valuable insights, they primarily rely on statistical approaches such as descriptive analysis or econometrics. Notably, the research outlined has not applied machine learning techniques to model the relationships between economic indicators and innovation. This research seeks to address this gap by applying the Random Forest Regressor and Multi-Layer Perceptron Neural Network architectures to predict a country's GII Rank using economic indicators.

III. Data and Method

Variables

Two main datasets were utilised; the GII data compiled by the World Intellectual Property Organisation (2024), and a dataset of global economic indicators capturing each country's economic conditions.

GII Ranks were based on a composite score combining Innovation Inputs (institutions, human capital, infrastructure) and Innovation Outputs (knowledge creation, creative outputs)

(WIPO, 2024). A complete collection of economic indicators utilised in this study, along with definitions of each, is available in Appendix 1.

Dataset Name	Contents	Source
Global Innovation Index	Ranks for the innovation performance of 149 unique countries between 2011 and 2024.	Kaggle (Kovacs, 2024)
Global Economy Indicators	Economic data from 1970 to 2021 for 220 unique countries, sourced from the United Nation's National Accounts Main Aggregates Database.	Kaggle (Patil, 2023)

Table 1: Overview of datasets used in the analysis.

Data Cleaning and Wrangling

The two datasets were merged using ISO alpha-3 country codes to ensure consistency and accuracy across country names. Rows with incomplete data were removed to retain only valid entries with complete data across all features. Columns with non-numeric data or missing values were removed. The GII "Rank" served as the dependent variable. From the Global Economy Indicators dataset, the economic variables were utilised as predictor variables. The final dataset included 1562 observations spanning the period from 2011 to 2021.

Model Selection and Justification

Two machine learning models were selected for this study: a Random Forest Regressor and a Multi-Layer Perceptron Neural Network, hereafter referred to as RFR and MLP, respectively.

Unlike decision trees, which can overfit to training data, RFRs use bootstrap aggregation and random feature selection to build multiple uncorrelated decision trees (Breiman, 2001). Then, the ensemble of trees is averaged through bagging to produce robust and generalisable predictions (Breiman, 2001). Furthermore, the RFR's feature importance metrics allow for interpretability and help identify which economic factors most strongly impact a country's innovation rank. Interpretability aligns well with the objective of producing insights that are policy relevant.

In addition, the MLP was employed to capture deeper, complex, and non-linear relationships between the predictor and target variables which traditional regressors may miss

(Goodfellow, Bengio, and Courville, 2016). The MLP architecture enabled the use of the complete feature set and to assess whether deeper models could enhance prediction.

Both models were trained and evaluated using a time-aware 80-20 train-test split. This ensured that data from years before 2019 was used for training, while later years were utilised for testing, to prevent data leakage. Country names which were categorical string variables were one-hot encoded to be included in the feature set.

IV. Interpretation and Discussion of Results

Exploratory Data Analysis

Exploratory data analysis was conducted to understand the structure, distribution, and interrelationships of the features used to predict GII Rank. Outliers were not removed because of the inequality present among global economies. Overall, the distributions of most variables were skewed, reflecting the concentration of economic prosperity among a small group of highly developed economies.

Correlation Analysis

Correlation analysis was conducted by calculating the Pearson coefficient between each predictor variable and the "Rank" column. Although a full correlation matrix was initially computed, the large number of variables made it difficult to interpret. Therefore, a Pearson correlation chart was used instead. The analysis also revealed multicollinearity among several predictor variables, which is common in social data science, as different aspects of the economy tend to influence one another.

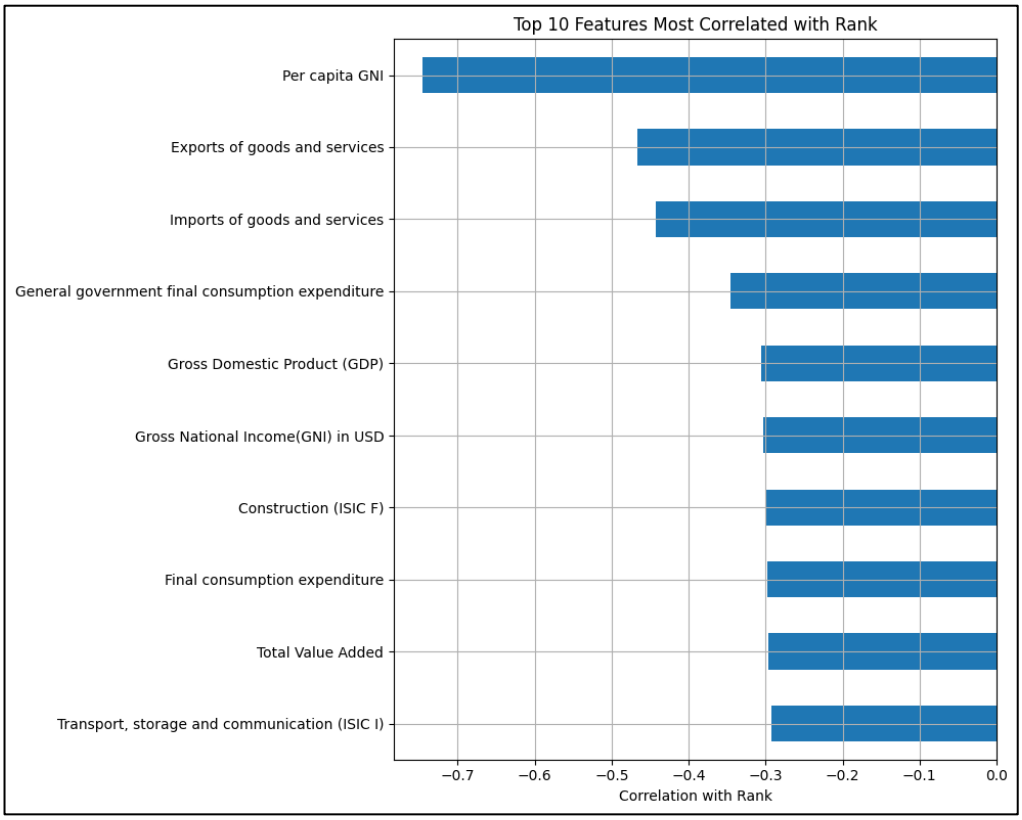


Figure 1: Top 20 Features Most Correlated with GII Rank

Scatter plots

Next, scatter plots of key indicators against the GII Rank column were generated to visually assess their relationships.

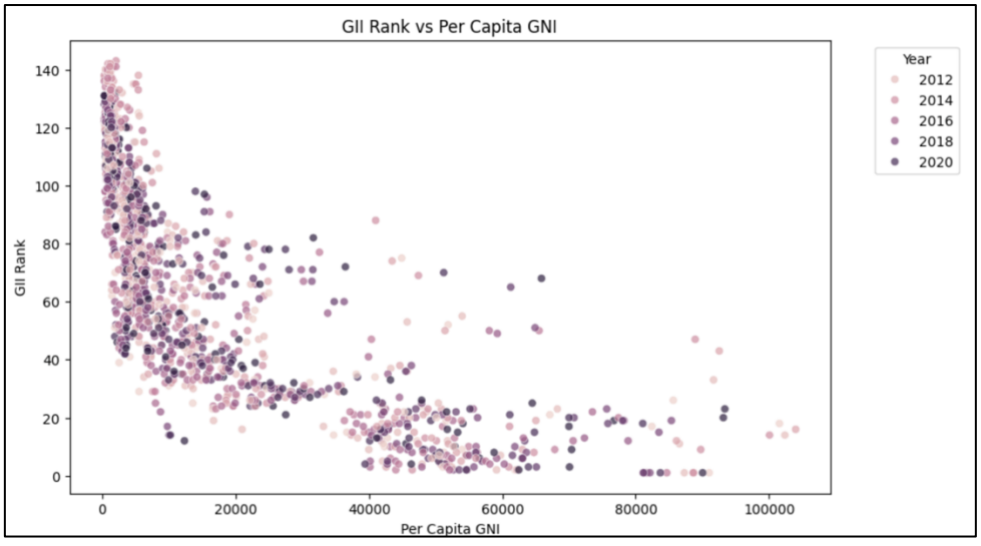


Figure 2: GII Rank vs Per capita GNI

Per capita Gross National Income (GNI), which exhibited the highest correlation with Rank, demonstrated a clear inverse relationship: as per capita GNI increased, the GII Rank tended to decrease numerically signifying a higher rank (Figure 2).

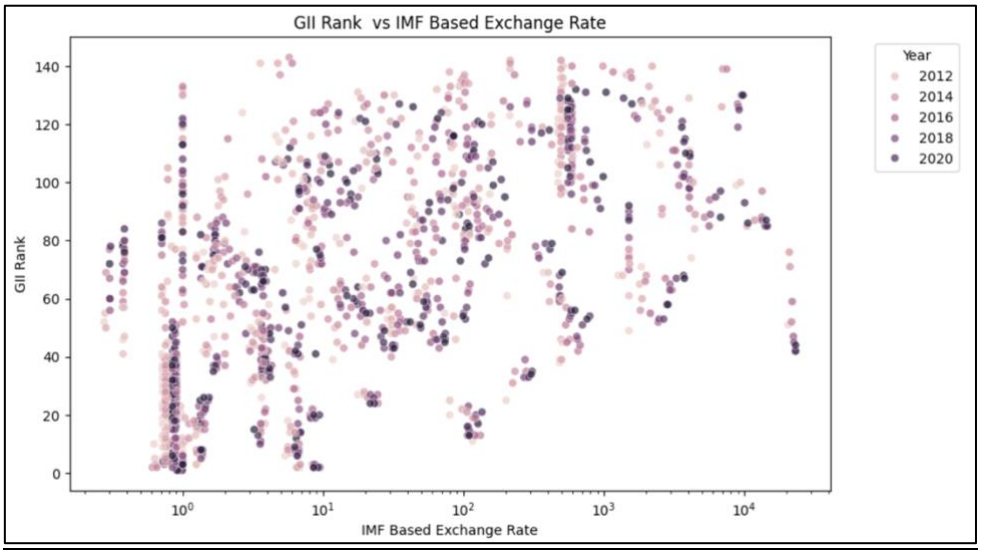


Figure 3: GII Rank vs IMF Based Exchange Rate (log scaled)

In contrast, the IMF-based exchange rate, which had a weaker correlation with Rank, showed no clear visual relationship in the scatter plot (Figure 3). To enhance interpretability and spread of the data, the IMF-based exchange rate values were log scaled.

Time-Series Line Plot

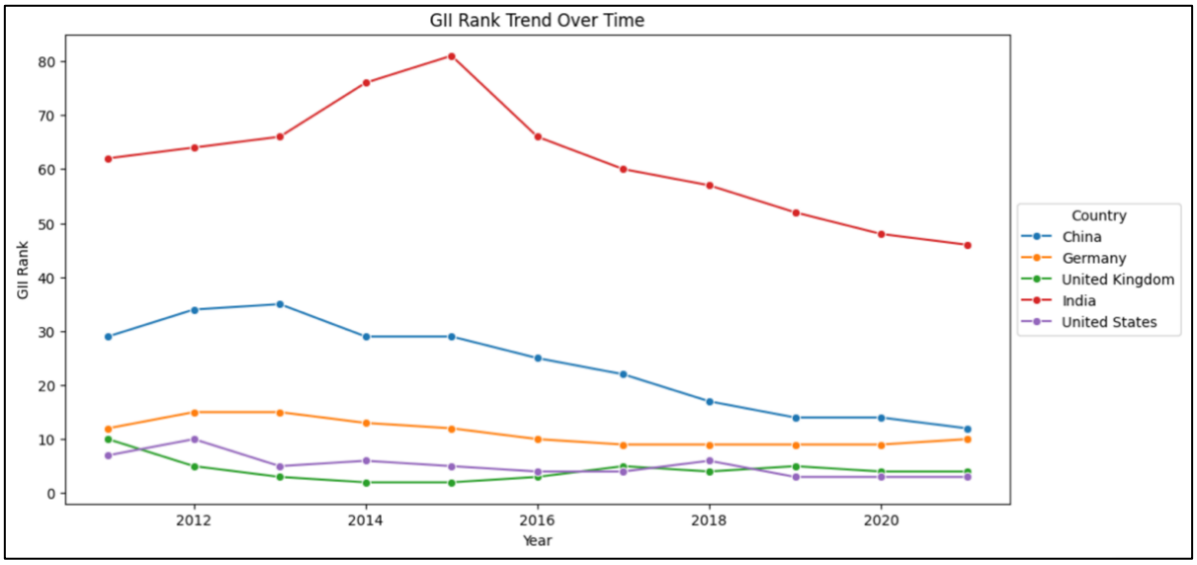


Figure 4: GII Rank Trend Over Time for Top Global Economies

To explore temporal trends, time-series plots of the GII Rank for the world's largest economies were generated. Countries like the US, Germany, and UK consistently maintained top 15 ranks (Figure 4). In contrast, India has shown steady improvement over time, while China experienced a notable rise in innovation rank, especially after 2015.

Model Results, Performance and Comparison

Model performance was evaluated using standard regression metrics including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2), with additional analysis of residuals and feature importances where applicable.

Model	MAE	RMSE	R^2
Random Forest Regressor	6.23	8.52	0.948
Multi-Layer Perceptron Neural Network	7.63	9.75	0.932

Table 2: Performance metrics of both models (3 significant figures)

The RFR outperformed the MLP across all evaluation metrics. The higher R^2 score (0.948 vs. 0.932) suggests that the RFR explained a greater proportion of variance in the GII Rank. Similarly, the lower RMSE (8.53 vs 9.75) further supports this. Finally, the lower MAE (6.24 vs. 7.64) demonstrates that RFR achieved smaller average prediction errors compared to the MLP. RMSE penalises larger errors more heavily than MAE, yet both were lower for the RFR suggests that it was more robust even against extreme prediction errors. Furthermore, the RFR also provided greater interpretability through its feature importance scores which could aid in policy-relevant insights (Figure 7).

These results align with findings from Shwartz-Ziv and Armon (2022), who show that deep learning models often underperform on tabular datasets. In such cases, tree-based methods like RFRs are better equipped, without the need for complex parameter tuning. The RFR's ensemble architecture, which uses bagging and feature randomness, allows it to generalise well and avoid overfitting, especially valuable in small to medium sized social science datasets. While the MLP captured deeper, non-linear patterns, it was ultimately less suited to this task due to the dataset's structure and size. Neural networks like MLPs require larger, high-dimensional datasets to reach peak performance, and they also do not provide transparent insights.

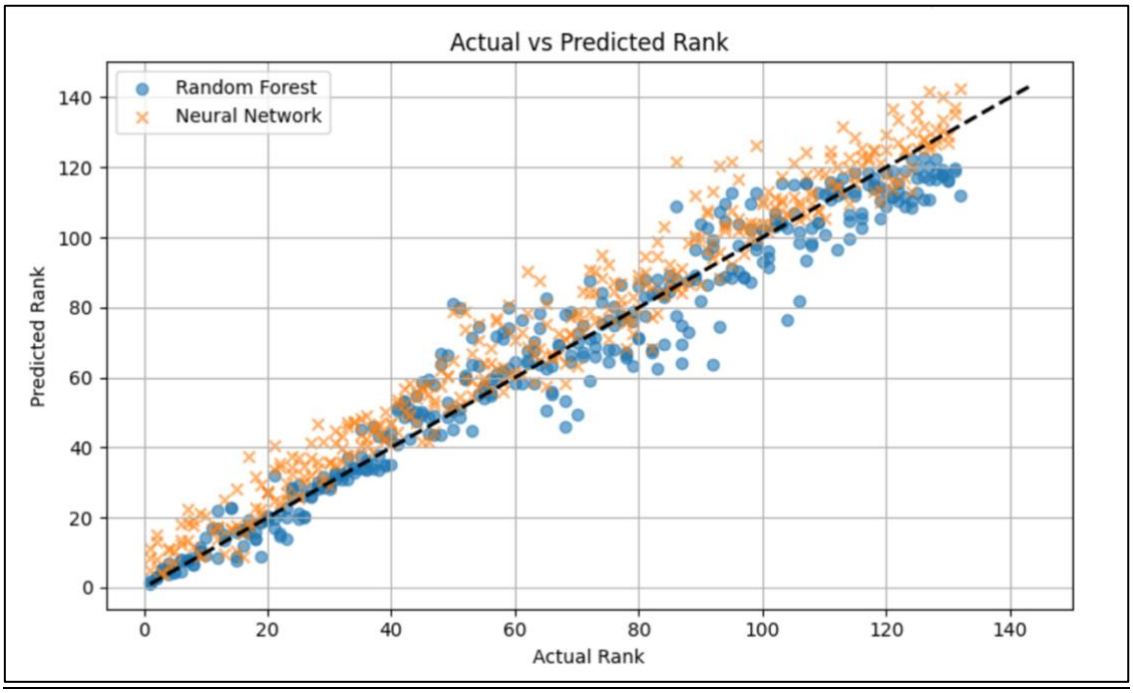


Figure 5: Actual vs Predicted Rank for the Random Forest Regressor and Multi-Layer Perceptron Neural Network Models

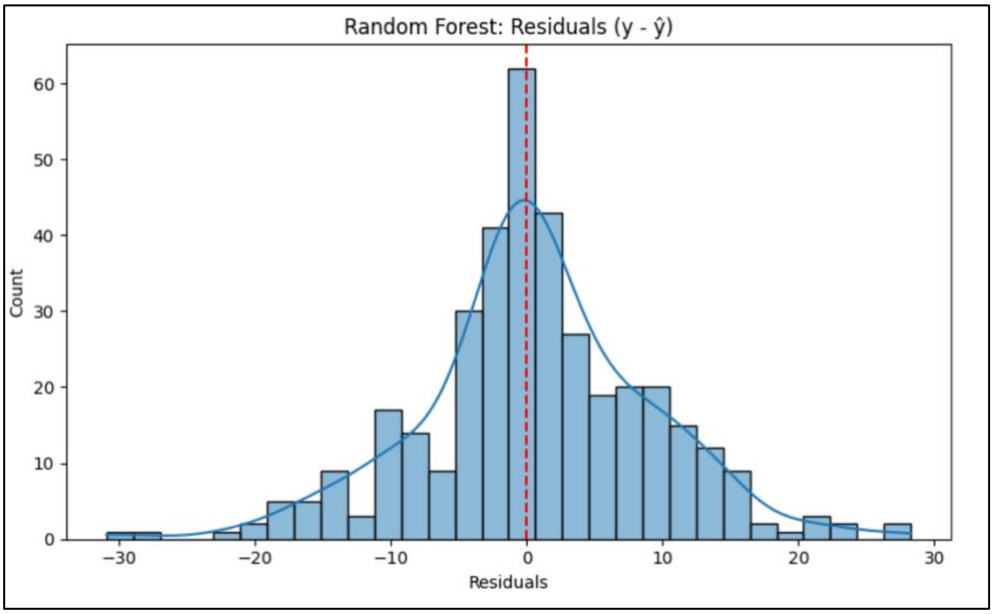


Figure 6: Random Forest Regressor Residuals

Residual analysis revealed that the RFR's residuals were symmetrically distributed and showed no signs of skewness or heteroscedasticity.

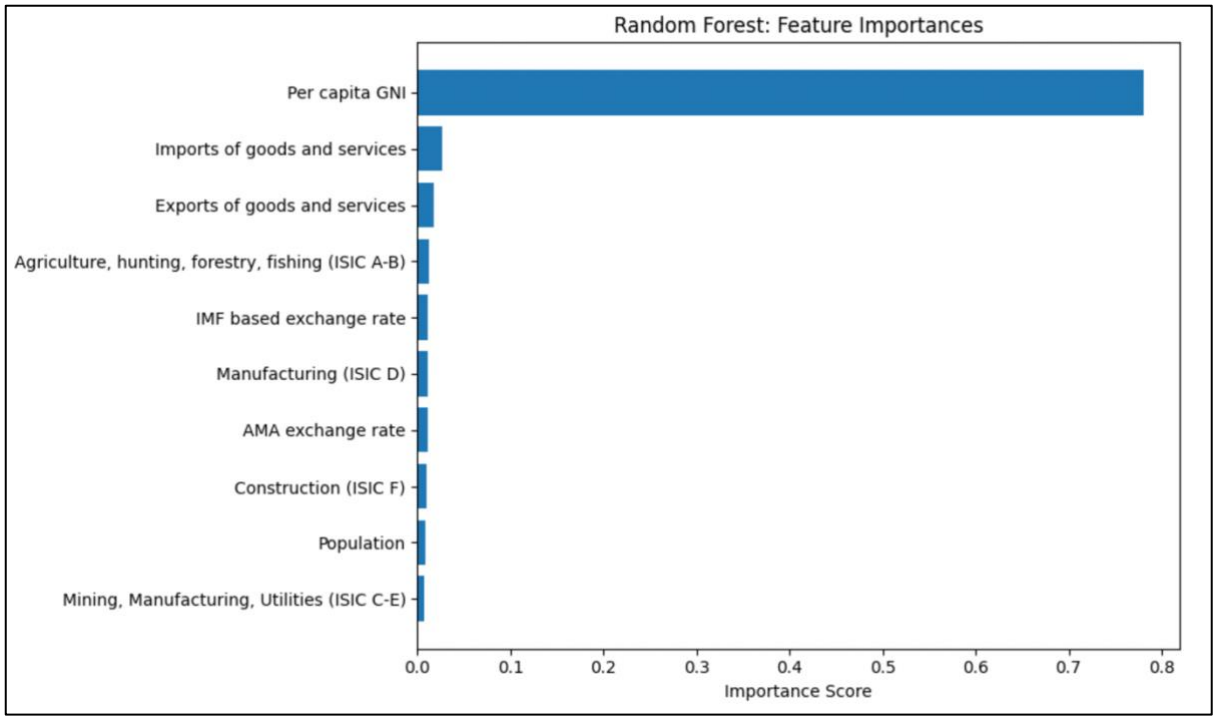


Figure 7: Random Forest Regressor Feature Importances

Feature importance analysis further confirmed the dominance of per capita GNI, which received a score close to 0.8. All other features contributed far less, each scoring below 0.1. This reinforces findings from exploratory and correlation analysis and suggests that per capita GNI is the strongest predictor of GII Ranks within this dataset.

Limitations and implications

These findings suggest a strong association between a per capita GNI and GII Rank. While causality cannot be established, the results imply that wealthier populations are more likely to benefit from the infrastructure, education, and R&D ecosystems that enable innovation. For policymakers aiming to improve GII Ranks, this highlights the potential of leveraging economic prosperity, particularly through inclusive and equitable income growth, as a foundation for innovation. Moreover, the low relative importance of trade indicators in the model suggests that internal economic strength may be more predictive of innovation than external trade alone, aligning with endogenous growth theory (Romer, 1994).

The main limitations of the approach included data availability and the opaque GII metric. Innovation can be driven by factors beyond the economy including political stability, education quality, and technological infrastructure, none of which were included in the dataset due to data

quality and availability. Furthermore, the GII is an index whose exact calculation methods are partially opaque, complicating the interpretation of model outputs (WIPO, 2024). This analysis takes a decomposition approach, meaning that rather than treating GII as a monolithic score, the analysis attempts to infer which underlying economic indicators contribute most heavily to its variation across countries.

Besides, the GII is not a purely objective measure; it aggregates subjective indicators such as survey responses and expert opinions, which introduces potential biases based on institutional or regional norms. As machine learning models are sensitive to biases in training data, the accuracy of predictions is constrained by the biases embedded in the GII itself. This limitation highlights the risk of treating innovation dynamics as a single quantifiable target without deeper causal understanding. Innovation rankings such as the GII may privilege economies with certain institutional forms, potentially disadvantaging alternative innovation models found in different political contexts.

V. Conclusion

This study found that national innovation ranks, measured by the Global Innovation Index, can be accurately predicted using a small set of economic indicators. The Random Forest Regressor achieved an R^2 of 0.948 and outperformed the Multi-Layer Perceptron across all metrics. However, the model's predictive power was concentrated in a single feature, per capita GNI, which dominated the feature importance analysis.

These results suggest that economic prosperity is a key enabler of national innovation. Countries with higher incomes likely have greater access to infrastructure, education, and R&D. For policymakers, this highlights the need to promote inclusive income growth and invest in innovation ecosystems.

Future research could strengthen this analysis by applying causal inference techniques to test whether higher per capita GNI drives innovation. Using more transparent and specific innovation metrics, such as patent filings, R&D expenditure, or startup rates, could also enhance clarity. Additionally, applying time-series or panel regression methods would better capture time-dependent dynamics across countries. Finally, expanding the dataset to include political, educational, social, and technological indicators could offer a more comprehensive view of what drives innovation across countries.

Appendices

Appendix 1

Indicator Name	Definition	Citation
AMA exchange rate	Average market exchange rate from the IMF's Annual Market Averages, used to compare international currencies.	IMF (n.d.)
IMF based exchange rate	Official IMF exchange, adjusted based on purchasing power for international comparisons.	IMF (n.d.)
Population	Total number of people living in a country, which impacts market size and labour force.	World Bank Group (n.d.)
Per capita GNI	Gross National Income per population of residents the economy.	World Bank Group (n.d.)
Agriculture, hunting, forestry, fishing	Value gained from the agriculture, hunting, forestry, and fishing industries.	UN National Accounts Main Aggregates Database (n.d.)
Construction	Value added from the construction sector, including infrastructure, and building development.	UN National Accounts Main Aggregates Database (n.d.)
Exports of goods and services	Total value of goods and services sold to global markets (as a percentage of the country's GDP).	World Bank Group (n.d.)
Final consumption expenditure	Total household and government payments on goods and services.	World Bank Group (n.d.)
General government final consumption expenditure	Government outflow on goods and services that directly satisfy the population's needs.	World Bank Group (n.d.)
Gross capital formation	Investment in fixed assets such as buildings, machinery, and infrastructure.	World Bank Group (n.d.)
Gross fixed capital formation (including Acquisitions less disposals of valuables)	Investment in fixed assets.	World Bank Group (n.d.)
Household consumption expenditure (including Non-profit institutions serving households)	Spending of households and non-profit organisations on goods and services.	World Bank Group (n.d.)
Imports of goods and services	Total worth of all goods and services bought from international markets.	World Bank Group (n.d.)

Table 3: Predictor variables

Indicator Name	Definition	Citation
Manufacturing	Value added from the manufacturing sector.	UN National Accounts Main Aggregates Database (n.d.)
Mining, Manufacturing, Utilities	Total value added from mining, manufacturing, and utilities sectors.	UN National Accounts Main Aggregates Database (n.d.)
Other Activities	Value added from miscellaneous sectors including financial services, education, and health.	UN National Accounts Main Aggregates Database (n.d.)
Total Value Added	Net output after deducting intermediate inputs.	World Bank Group (n.d.)
Transport, storage and communication	Value added from transport, logistics, postal, and telecommunication services.	UN National Accounts Main Aggregates Database (n.d.)
Wholesale, retail trade, restaurants and hotels	Value added from wholesale, retail, food service, and accommodation sectors.	UN National Accounts Main Aggregates Database (n.d.)
Gross National Income (GNI) in USD	Sum of value added by residents plus, income from nationals abroad.	World Bank Group (n.d.)
Gross Domestic Product (GDP)	Total market value of all complete goods and services produced within a country.	World Bank Group (n.d.)

Table 3 (Continued): Predictor variables

Reference List

Alfaro, L., Cuñat, A., Fadinger, H. and Liu, Y. (2023). 'The Real Exchange Rate, Innovation and Productivity', *Journal of the European Economic Association*, 21(2), pp.637-689. doi: 10.1093/jeea/jvac058.

Bloom, N., Reenen, J.V. and Williams, H. (2019). 'A Toolkit of Policies to Promote Innovation', *Journal of* Economic Perspectives, 33(3), pp.163–84. doi: 10.1257/jep.33.3.163.

Bramer, M. (2013). 'Avoiding Overfitting of Decision Trees', in *Principles of Data Mining, Undergraduate Topics in Computer Science*. Springer, London. doi: 10.1007/978-1-4471-4884-5_9.

Breiman, L. (2001). 'Random forests', *Machine Learning*, 45(1), pp.5–32. doi: 10.1023/A:1010933404324.

Chan, K.Y., Abu-Salih, B., Qaddoura, R., Al-Zoubi, A.M., Palade, V., Pham, D.S., Del Ser, J. and Muhammad, K. (2023). 'Deep neural networks in the cloud: Review, applications, challenges and research directions', *Neurocomputing*, 545. doi: 10.1016/j.neucom.2023.126327.

Damanpour, F. and Schneider, M. (2006). 'Phases of the Adoption of Innovation in Organizations: Effects of Environment, Organization and Top Managers', *British Journal of Management*, 17, pp.215-236. doi: 10.1111/j.1467-8551.2006.00498.x.

Deleidi, M., and Mazzucato, M. (2019). 'Putting Austerity to Bed: Technical Progress, Aggregate Demand and the Supermultiplier', *Review of Political Economy*, 31(3), pp.315–335. doi: 10.1080/09538259.2019.1687146.

Deleidi, M., De Lipsis, V., Mazzucato, M., Ryan-Collins, J. and Agnolucci, P. (2019). *The macroeconomic impact of government innovation policies: A quantitative assessment.* UCL Institute for Innovation and Public Purpose, Policy Report (IIPP 2019-06). Available at: https://www.ucl.ac.uk/bartlett/public-purpose/wp2019-06 (Accessed: 23 April 2025).

Ding, H. (2022). 'What kinds of countries have better innovation performance?–A country-level fsQCA and NCA study', *Journal of Innovation & Knowledge*, 7(4). doi: 10.1016/j.jik.2022.100215.

Furman, J.L., Porter, M.E. and Stern, S. (2002). 'The Determinants of National Innovative Capacity', *Research Policy*, 31(6), pp.899-933. doi: 10.1016/S0048-7333(01)00152-4.

Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep learning*. MIT Press: Cambridge, MA. Available at: http://www.deeplearningbook.org (Accessed: 23 April 2025).

Hsu, P.H., Taylor, M.P., Wang, Z. and Xu, Q. (2022). 'Currency volatility and global technological innovation', *Journal of International Economics*, 137. doi: 10.1016/j.jinteco.2022.103607.

IMF (n.d.) International Financial Statistics Database. Available at: https://data.imf.org/ (Accessed: 23 April 2025).

Ivanova, I., Strand, O., Kushnir, D. and Leydesdorff, L. (2017). 'Economic and technological complexity: A model study of indicators of knowledge-based innovation systems', *Technological Forecasting and Social Change,* 120, pp.77-89. doi: 10.1016/j.techfore.2017.04.007.

Kovács, K. (2024). 'Global Innovation Index (WIPO) - 2011-2024 Data'. Available at: https://www.kaggle.com/datasets/karlakovacs/global-innovation-index-wipo-2011-2024-data (Accessed: 23 April 2025).

Malik, S. (2020). 'Macroeconomic Determinants of Innovation: Evidence from Asian Countries', *Global Business Review*, 24(1), pp.137-151. doi: 10.1177/0972150919885494.

Mazzucato, M. (2016). 'From market fixing to market-creating: a new framework for innovation policy', *Industry and Innovation*, *23*(2), pp.140–156. doi: 10.1080/13662716.2016.1146124.

Mazzucato, M. and Dibb, G. (2019). *Missions: A beginner's guide*. UCL Institute for Innovation and Public Purpose, Policy Brief series (IIPP PB 09). Available at: https://www.ucl.ac.uk/bartlett/publications/2019/dec/missions-beginners-guide (Accessed: 23 April 2025).

Mowery, D.C. (1998). 'The changing structure of the US national innovation system: implications for international conflict and cooperation in R&D policy', *Research Policy*, 27(6), pp.639-654. doi: 10.1016/S0048-7333(98)00060-2.

Nasir, M.H. and Zhang, S. (2024). 'Evaluating innovative factors of the global innovation index: A panel data approach', *Innovation and Green Development*, 3(1). doi: 10.1016/j.igd.2023.100096.

OECD (2011). 'Demand-side Innovation Policies', OECD Publishing. doi: 10.1787/9789264098886-en.

OECD (2023). 'OECD Science, Technology and Innovation Outlook 2023: Enabling Transitions in Times of Disruption', OECD Publishing: Paris. doi: 10.1787/0b55736e-en.

Patil, P. (2023), 'Global Economy Indicators'. Available at: https://www.kaggle.com/datasets/prasad22/global-economy-indicators (Accessed: 23 April 2025).

Raghupathi, V. and Raghupathi, W. (2017). 'Innovation at country-level: association between economic development and patents', *J Innov Entrep*, 6(4). doi: 10.1186/s13731-017-0065-0.

Romer, P.M. (1994). The Origins of Endogenous Growth. *The Journal of Economic Perspectives*, *8*(1), 3–22. http://www.jstor.org/stable/2138148.

Shabbir, M.N., Liyong, W. and Iftikhar, K. (2021). 'Trade Openness, innovation, and Economic Growth: A causal Effect Analysis of OECD Countries', *Pakistan Journal of Economic Studies (PJES)*, *4*(1), pp.63–98. Available at: https://www.researchgate.net/publication/353237490_Trade_Openness_Innovation_and_Economic_Growth_A_causal_Effect_Analysis_of_OECD_and_Non-OECD_Countries (Accessed: 23 April 2025).

Shwartz-Ziv, R. and Armon, A. (2022). 'Tabular data: Deep learning is not all you need', *Information Fusion*, 81, pp.84-90. doi: 10.1016/j.inffus.2021.11.011.

Taylor, S.P. (2017). 'What is innovation? A study of the definitions, academic models and applicability of innovation to an example of social housing in England', *Open Journal of Social Sciences*, 5(11). pp.128-146. doi: 10.4236/jss.2017.511010.

UN National Accounts Main Aggregates Database (n.d.). Available at: https://unstats.un.org/unsd/nationalaccount/data.asp (Accessed: 23 April 2025).

United Kingdom. Department for Business, Energy and Industrial Strategy (2021). *UK Innovation Strategy. Leading the future by creating it.* Available at: https://assets.publishing.service.gov.uk/media/61110f2fd3bf7f04402446a8/uk-innovation-strategy.pdf (Accessed: 23 April 2025).

World Bank Group (n.d.). *Metadata Glossary.* Available at: https://databank.worldbank.org/metadataglossary/world-development-indicators/series/NE.GDI.TOTL.ZS#:~:text=Gross%20capital%20formation%20(formerly%20gross,in%20the%20level%20of%20inventories (Accessed: 17 October 2022).

World Intellectual Property Organisation (WIPO). (2024). 'Global Innovation Index 2024 Unlocking the Promise of Social Entrepreneurship', Geneva: WIPO. doi: 10.34667/tind.50062.

Zhu, Z., Morrison, G., Puliga, M., Chessa, A. and Riccaboni, M. (2018). 'The similarity of global value chains: A network-based measure', *Network Science*, 6(4), pp.607-632. doi: 10.1017/nws.2018.8.